Fine-tuning Whisper for Air Traffic Control Speech with Limited Training Data

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Abstract

Air traffic control (ATC) radio is fast, accented and band-limited, and general speech recognisers handle it poorly. Starting from one hour of public ATC audio (874 clips from the ATCO2 corpus), I fine-tuned Whisper, later adding 10.5 hours of real ATC speech from the UWB-ATCC corpus. An audit of my 2025 prototype, which had reported 20.31% word error rate (WER), found that 101 of its 175 test clips shared recordings with training clips and that the test clips had been used for tuning. On 74 clips from recordings no model trained on, unmodified Whisper-small scores 56.87% and the 2025 model 18.64%. The best model, an average of two Whisper-medium models trained with low-rank adapters, scores 14.27%, 4.37 points better than the 2025 model (95% interval 2.07 to 6.76); most of the gain comes from the larger model. Checkpoints and decoding were chosen on a separate validation set and the final run followed a rule fixed in advance, but some later ideas came after earlier test results were known, so the test set is not an untouched holdout. For the Whisper-small recipe, cross-validation by recording gives 19.16% and leaving out each airport in turn 26.48%. Synthetic radio speech did not help in these experiments. At the level of instructions, the word tagger extracts the same callsign, commands and numbers from the best model's transcript as from the correct one in only 38 of 74 clips.

1. Introduction

Controllers and pilots speak a restricted form of English. Numbers are read digit by digit, nine is said "niner", and most calls follow the same pattern: who is being addressed, what they must do and a value. The speech is fast, spoken in every accent and sent over amplitude-modulated radio whose narrow audio channel loses the lowest and highest frequencies; ICAO's guidance for 8.33 kHz channels assumes about 2.5 kHz of audio bandwidth [11], narrower than the 300 Hz to 3.4 kHz telephone band. General-purpose speech recognisers, trained mostly on other kinds of speech, struggle with it; in early tests Whisper-small turned the callsign "Eurotrans" into "year trans" and wrote whole sentences from pure static.

Automatic transcription could help controllers, for example by filling in radar labels from their own speech, which has been shown to reduce wrong or missing entries in a controlled trial [8]. That requires the recogniser to be accurate on the words that matter, and it requires a trustworthy estimate of that accuracy on speech the system has never heard. This project began with the first question and ended mostly on the second.

The work has two parts. In 2025 I fine-tuned Whisper-small [1] on one hour of ATC audio and built a word tagger that labels callsigns, commands, values and waypoints, as separate scripts. A review in 2026 found that the evaluation did not measure what it appeared to, and the evaluation was rebuilt, the models retrained, and the parts joined into one application. This paper reports the rebuilt evaluation. Its contributions are:

1. A test set of recordings unseen by every model, with all development choices made on a separate validation set, and paired bootstrap intervals for every comparison.
2. Measured answers to questions the 2025 work left open: how much fine-tuning helps, whether a label bug mattered, whether synthetic radio speech helps, and whether more real speech or a larger model helps.
3. Five-fold cross-validation by recording, so that every clip is scored by a model that never heard its recording.
4. A measurement of how quickly accuracy falls as noise is added, with and without a voice-band filter.
5. A measurement of how transcription errors propagate into the extracted instruction fields.

2. Data

2.1 ATCO2

All evaluation uses the free one-hour subset of the ATCO2 corpus (ATCO2-test-set-1h) [2], released for research: short clips of controller and pilot speech from several European airports and Sydney, with transcripts. The subset was published as a test set; this project splits it into its own training and evaluation parts, so no score here can be compared with published ATCO2 results. Transcripts are lower case with digits spelled out.

The 2025 split divided the 874 clips into 699 for training and 175 for testing, clip by clip. A recording in ATCO2 is cut into several clips, and 101 of the 175 test clips came from recordings that also supplied training clips (92 recordings in all), so the model had already heard the same voices, frequencies and background noise. The other 74 test clips come from 66 recordings with no training clips.

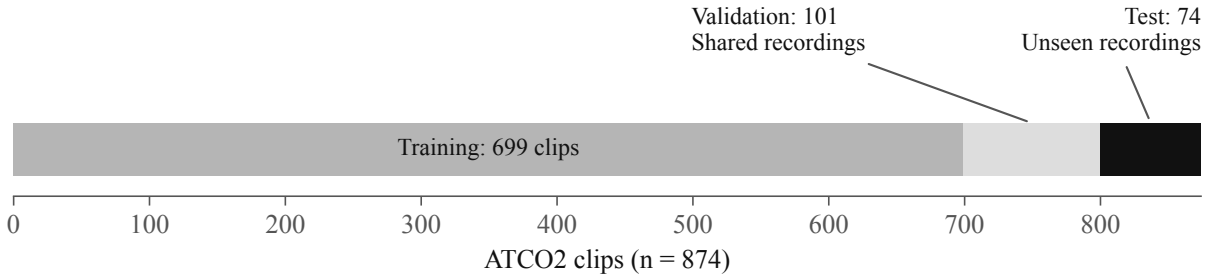


Figure 1. The split used in this study. The 2025 test set combined the validation and test clips shown here.

The 2026 evaluation keeps the 699 training clips unchanged and divides the old test clips by recording (Figure 1). The 101 clips from recordings that also supplied training clips become a validation set, used for every choice of checkpoint, decoding setting and model. The 74 clips from unseen recordings become the test set, used only to report results. Because the 2025 model was trained on exactly the same 699 clips, it can be scored on the same test set, which makes old and new models directly comparable. One caveat applies: the 2025 checkpoint itself was chosen in 2025 using the old test set, which includes these 74 clips, so its score may be slightly optimistic.

2.2 UWB-ATCC

The UWB-ATCC corpus [3] contains hand-transcribed controller and pilot speech from Czech airspace, released under CC BY-NC-SA 4.0. Its training split, 11,291 clips and 10.5 hours, is used only for training in this study. Its transcripts follow nearly the same conventions as ATCO2; four spellings were mapped to the ATCO2 form (decimal to point, xray to x ray, alfa to alpha, ok to okay), a mapping decided from the ATCO2 training and validation transcripts only.

2.3 Synthetic radio speech

In 2025 I generated 5,000 sentences from templates that follow ATC phraseology, with values drawn only from valid ranges (for example, transponder codes are octal, so no digit 8 or 9 appears). Microsoft Edge text-to-speech read them in 31 English voices at random speaking rates, and a simulated radio channel degraded the audio: band-limiting to 4 kHz, low-frequency engine noise with a $1/f^2$ spectrum (white noise summed over time), broadband static, μ-law companding, a 300 to 3,400 Hz band-pass filter, clipping and a squelch burst. A 2026 simulation of the script's noise stages (2,000 random clips) shows that the engine noise barely survived: about 99% of its power lies below 20 Hz, the later band-pass keeps a median 0.012% of it, and after the filter it sits a median 34 dB below the static. The training audio therefore contained almost no engine noise. 4,808 of the clips are in the training list used here. Two flaws affect the 2026 test of this data (run B). The airline list included callsigns I had picked from errors on the old test clips, which can only have favoured

run B. And its transcripts write "decimal" and "x-ray" where ATCO2 writes "point" and "x ray", and were not converted; run B wrote "decimal" once on the test clips, so this cost it at most a word or two.

3. Auditing the 2025 work

The 2025 model was re-scored without retraining. On the 175 old test clips it scores 20.88% WER with greedy decoding (435 word edits across 2,083 reference words) and 20.31% with beam search (423 edits). Four problems emerged when the scripts and numbers were re-read.

The test was not unseen. As described in Section 2.1, 101 of the 175 test clips shared a recording with training clips. The test clips had also been used to pick checkpoints, tune decoding and choose the "hard words" around which extra synthetic sentences were generated.

The labels did not match inference. The 2025 training labels began with the start-of-transcript token followed by the no-timestamps token, and the data collator prepended a second start token. At inference, however, Whisper is prompted with start-of-transcript, English, transcribe and no-timestamps. The model was therefore trained on a slightly different task from the one it performed.

One scorer changed its targets. A 2025 evaluation script reached about 19% by applying phrase replacements chosen from observed errors to both predictions and references, and by deleting substrings such as NE and JA wherever they appeared, which turns ONE NINE JAPAN into O NI PAN. Reproducing it on the beam output gives 386 edits over 2,022 words, or 19.09%. That is a different scoring target, not a better transcription, and it is not used.

The tagger was marking its own homework. The word tagger's reference labels came from word-list rules, not from people. The rule for waypoints labels any unfamiliar word of three or more letters as a waypoint, which is far too loose.

Other historical numbers are set aside for similar reasons. A comment in a 2025 script records values of 76.18, 76.23 and 76.67 after three training rounds, labelled "acc"; the scripts computed that as 100 minus WER, but after a clean-up that altered the text, with a decoding prompt built from test-set errors, and on a test file that cannot now be confirmed. A 2025 experiment with Whisper-large-v3 in 4-bit precision with LoRA adapters scored 27.86% on the old test clips, under training and decoding settings that differ from every other model here.

4. Methods

4.1 Models

All speech models are fine-tuned from the public Whisper checkpoints [1]. Whisper-small (242 million parameters) is fine-tuned in full. Whisper-medium (764 million parameters) is too large to fine-tune in full in the memory of the laptop used here (an Apple M3 Pro with 18 GB of shared memory), so its weights were frozen and rank-32 LoRA adapters [4] were trained on every attention and feed-forward projection: 34.6 million trainable parameters, 4.3% of the model. After training the adapters are merged into the weights, so the result is an ordinary checkpoint.

A model soup [5] averages the weights of models fine-tuned from the same starting point; here it is applied to pairs of Whisper-small models and to the two Whisper-medium models. Such models often lie in the same basin of the loss surface, so their average can outperform each of them without any extra inference cost.

4.2 Training

All runs use the corrected labels, whose prefix matches the inference prompt, with the start token appearing once. Other settings: AdamW with weight decay 0.01, a linear learning-rate schedule with 50 warm-up steps, an effective batch of 16 clips, gradient clipping at 1.0, and SpecAugment [6] with time and frequency masking probabilities of 0.05. The learning rate is 1×10^{-5} for Whisper-small and 5×10^{-4} for the LoRA adapters. After every epoch the model transcribes the validation clips with greedy decoding and repetition guards; the checkpoint with the lowest validation WER is kept. Each run had a fixed epoch budget (12 for run A, 8 for runs B and E and for each cross-validation fold, 6 for the Whisper-medium run with extra speech and 5 for

the ATCO2-only one), with the learning rate falling to zero at the end of it, and would stop early if validation WER did not improve for two to four epochs. Only the runs that never beat their starting point stopped early. Run E and both Whisper-medium runs were still improving at their final epoch, so they may be undertrained; a repeat of the Whisper-medium run with 12 planned epochs (Section 5.1) peaked at epoch 9.

All 2026 training ran on the M3 Pro laptop's GPU; the 2025 models had been trained on a Windows laptop with an NVIDIA RTX 3070, which I no longer have. Three measures kept memory at about 6 GB for Whisper-small and 4.9 GB for Whisper-medium: gradient checkpointing, one clip per step with gradients accumulated over 16, and labels padded to a single fixed length. The last matters on Apple GPUs, where every new tensor shape compiles and caches another compute graph; with variable label lengths, memory grew until the run failed. Whisper-small trained at about 0.6 seconds per clip and Whisper-medium at about 2.9.

Where extra data was used, a fresh random sample of extra clips was drawn each epoch and mixed with all 699 real ATCO2 clips: 700 synthetic clips per epoch, 1,400 UWB-ATCC clips per epoch for Whisper-small, and 1,000 for Whisper-medium. With UWB-ATCC, the ATCO2 clips were therefore outnumbered in every epoch.

4.3 Decoding

Three decoding settings were scored for each model: greedy decoding; beam search with five beams, a repetition penalty of 1.2 and no repeated three-token sequences (the 2025 settings, called guarded beam search here); and plain beam search with five beams. For each model, the setting with the lowest validation WER is the one reported. Plain greedy decoding occasionally falls into a loop on noisy or non-English speech, repeating a phrase until the token limit; one such clip added 68 errors to a validation score during development, which is why the guards are also used whenever a model is checked during training.

4.4 Evaluation

WER is the number of substituted, deleted and inserted words divided by the number of reference words, pooled over all clips: $\text{WER} = (S + D + I)/N$. Before scoring, text is converted to upper case, punctuation is removed, and digits are spelled out, because unmodified Whisper writes "4402" where the references say "four four zero two". Other formatting differences, such as "X-ray" against "x ray", are left in place and slightly penalise the unmodified model.

Checkpoints and decoding settings were selected on validation, and the final longer run followed a rule written beforehand; some later experiment ideas, notably averaging the two Whisper-medium models, were developed after earlier test results were known. The test clips are therefore disjoint from the training data but not an untouched final holdout, and the intervals below measure sampling uncertainty, not any bias that adapting the investigation could introduce.

Uncertainty is estimated by resampling the 74 test clips with replacement 10,000 times [7]. Differences between two models use the same resampled clips for both (a paired bootstrap), which removes much of the variation due to which clips happen to be hard.

Two further measurements address the practical question. For **noise robustness**, white Gaussian noise was added to each test clip at a signal-to-noise ratio $\text{SNR} = 10 \log_{10}(P_{\text{signal}}/P_{\text{noise}})$ of 30, 20, 15, 10, 5 and 0 dB, where P_{signal} is the mean power of the whole clip, pauses included, after the optional band-pass filter (so at 0 dB the speech itself is somewhat louder than the noise, which pauses pull down), with a fixed random seed per clip so that all models hear identical audio; decoding was greedy with repetition guards. For **instruction-level agreement**, the application's DistilBERT tagger was run on the correct transcript and on each model's transcript, and the extracted (field, words) pairs were compared clip by clip. This measures how much transcription error reaches the extracted fields; it does not measure the tagger's own accuracy.

5. Results

5.1 Recognition on unseen recordings

Table 1. Word error rate on the 101 validation clips and the 74 test clips. Each model uses the decoding setting that did best on validation. Intervals are 95% bootstrap intervals over test clips.

Model	Validation WER	Test WER	95% interval
Whisper-small, not fine-tuned	63.81%	56.87%	49.53 to 64.81
Whisper-medium, not fine-tuned	47.20%	43.88%	38.09 to 50.11
2025 model	21.68%	18.64%	14.44 to 23.19
Retrained with the corrected labels	20.02%	19.28%	15.37 to 23.35
As above, plus synthetic radio speech	23.08%	20.34%	16.05 to 25.00
As above, plus UWB-ATCC	21.68%	18.21%	13.78 to 23.48
Soup of the corrected and UWB-ATCC models	17.66%	16.72%	13.00 to 20.68
Whisper-medium with LoRA, plus UWB-ATCC	16.61%	15.23%	11.69 to 19.13
Ablation: Whisper-medium with LoRA, ATCO2 only	17.66%	15.65%	12.38 to 19.17
Soup of the two Whisper-medium models	14.86%	14.38%	10.95 to 18.17
The same soup, first model trained for 12 epochs	14.42%	14.27%	10.93 to 17.99

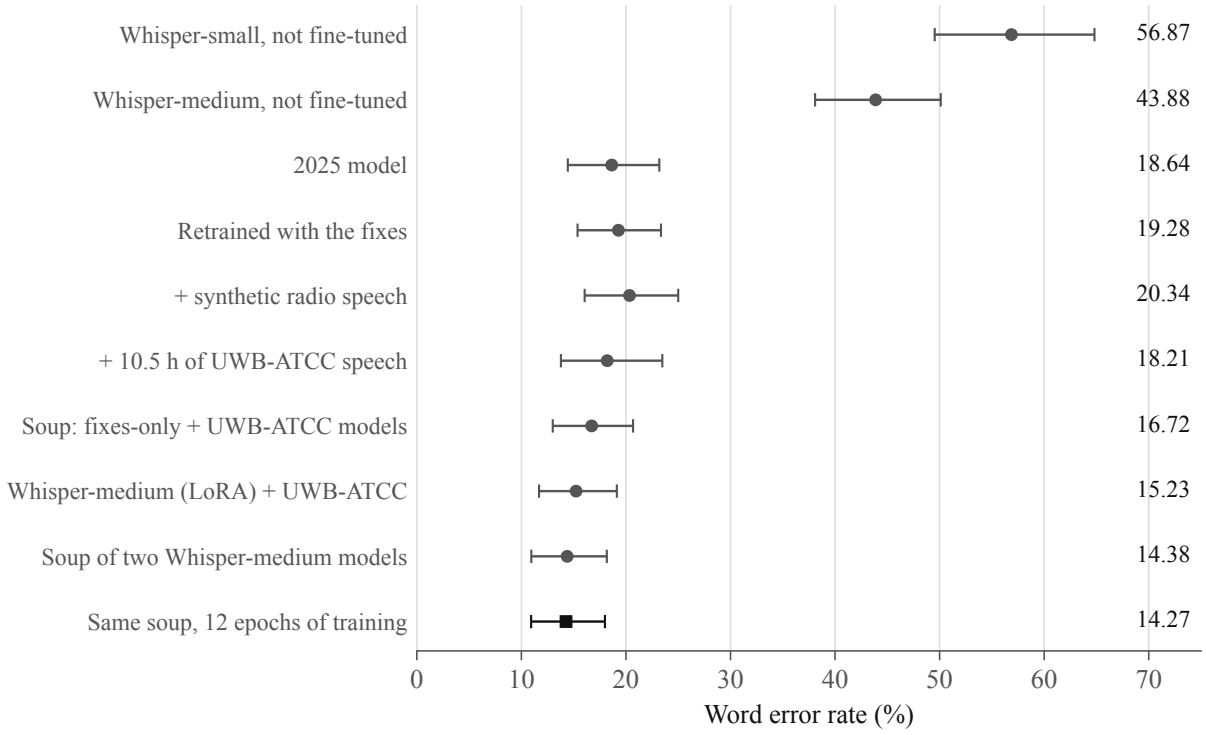


Figure 2. Word error rate on the 74 test clips, with 95% bootstrap intervals. The square identifies the model selected on validation.

Fine-tuning is by far the largest effect: from 56.87% for the unmodified model to 18.64% for the 2025 model on the same clips. Model size alone does much less: unmodified Whisper-medium scores 43.88% (95% interval 38.09 to 50.11), so most of the way from there to the best model's 14.27% is adaptation to ATC speech. The 2025 model does slightly better on the unseen test clips than on all 175 old test clips (20.31%), so any inflation of its earlier score by the recording overlap is too small to detect this way.

The label correction did not change accuracy measurably. Synthetic speech did not help, even in a controlled comparison on unseen recordings. Adding 10.5 hours of real speech changed the test score little, but

Table 2. Change in test WER relative to the 2025 model, in percentage points, with paired 95% intervals.

Model	Change	95% interval
Retrained with the corrected labels	+0.64	−2.13 to +3.39
Plus synthetic radio speech	+1.70	−1.29 to +4.87
Plus UWB-ATCC	−0.43	−3.13 to +2.42
Soup	−1.92	−4.54 to +0.75
Whisper-medium with LoRA, plus UWB-ATCC	−3.41	−5.77 to −1.08
Ablation: Whisper-medium with LoRA, ATCO2 only	−2.98	−5.26 to −0.67
Soup of the two Whisper-medium models	−4.26	−6.61 to −1.96
The same soup, first model trained for 12 epochs	−4.37	−6.76 to −2.07

it made the model far less prone to looping: under plain greedy decoding its test WER was 19.38% against 26.52% for the model without it. Averaging the two models' weights gave the lowest validation error among the Whisper-small models, and a test score 1.92 points below the 2025 model, a gain whose interval still includes zero. Continuing to train the 2025 checkpoint with the corrected labels never improved on it on validation, so no such model was kept.

Whisper-medium with LoRA adapters had a lower validation error than any Whisper-small model. On the test clips it makes 143 word errors against 175 for the 2025 model, a reduction of 3.41 points whose 95% interval (1.08 to 5.77 points) excludes zero.

To separate model size from the extra data, an ablation declared in advance, and not eligible for selection, trained Whisper-medium in the same way on the 699 ATCO2 clips alone. It scores 17.66% on validation and 15.65% on the test clips (147 word errors), 2.98 points better than the 2025 model (95% interval 0.67 to 5.26 points). Most of the gain therefore comes from the larger model; adding UWB-ATCC improved on the ablation by a further 0.42 points on test and 1.05 on validation, differences too small to establish here. The ablation trained for five epochs of 699 clips rather than six of 1,699, so it also saw less audio.

Averaging the weights of the two Whisper-medium models gave the lowest validation error at that point, 14.86%, and 135 word errors on the test clips: 14.38%, 4.26 points better than the 2025 model (95% interval 1.96 to 6.61 points). This choice was not blind: the ATCO2-only model had been declared an ablation that could not itself be selected, and I built the average after both Whisper-medium models had been scored on the test clips. Restarting the single Whisper-medium model with fresh adapters and a new warm-up for up to three more epochs did not help: its validation error rose from 17.92% to 18.09% and 18.44% (greedy decoding with guards). Because the kept runs had all been improving at their last epoch, the Whisper-medium run was then repeated from the start with 12 planned epochs instead of 6, under a rule written before it ran: the better of the new model and its average with the ATCO2-only model would be adopted only if it made fewer than 170 word errors on validation. On its own the 12-epoch model scores 15.65% on validation against 16.61% for the 6-epoch one, with its best epoch at 9. Its average made 165 errors (14.42%), so it was adopted and scored once on test: 134 word errors, 14.27%, 4.37 points better than the 2025 model (95% interval 2.07 to 6.76 points). Against the 6-epoch soup it is one word better (−0.11 points, 95% interval −1.56 to +1.50), so the two are level on the evidence available; it is the selected model because the rule chose it on validation.

5.2 Every recording and every airport

The 74-clip test set is small, so the Whisper-small recipe was also evaluated by five-fold cross-validation over all 874 clips. The clips were grouped by recording and the recordings dealt into five folds of 174 to 177 clips; for each fold a fresh model was trained on the other four, minus a tenth of their recordings held out as validation for choosing the checkpoint, and then scored on the fold. Every clip is therefore transcribed by a model that never trained on its recording. Decoding was fixed in advance to guarded beam search.

The pooled WER of 19.16% agrees with the 74-clip estimate for the same recipe (19.28%) and has an interval about a third as wide. The folds range from 17.74% to 21.01%, a spread that shows how much a single small test set depends on which recordings it happens to contain.

Table 3. Five-fold cross-validation by recording over all 874 clips (10,930 reference words).

Model	Pooled WER	95% interval
Whisper-small, not fine-tuned	53.44%	51.13 to 55.81
Whisper-small retrained with the corrected labels	19.16%	17.90 to 20.45

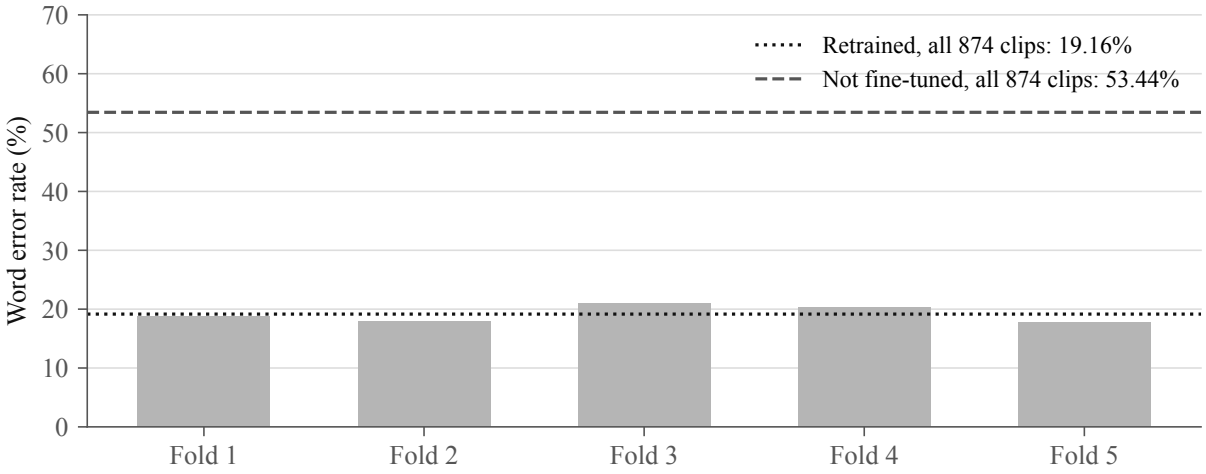


Figure 3. Word error rate of each cross-validation fold, and the pooled rate over all 874 clips.

The cross-validation still lets each model hear other recordings from the same airport: the same controllers, frequencies and place names. A stricter test, declared before it ran, left out one airport at a time. The same recipe was trained seven times, each time without one of the seven airports, with a tenth of the remaining recordings held out to choose the checkpoint, and scored on every clip from the airport it never heard. Pooled over all 874 clips it scores 26.48% (95% interval 24.96 to 27.99%), 7.32 points worse than the cross-validation by recording on the same clips (paired 95% interval 6.32 to 8.36), and still about half the unmodified model's 53.44% (Table 4). The cost depends on the airport: little at Brno and Bern, about 12 points at Sion, and 23 at Sydney, the only airport outside Europe, whose model had heard no Australian speech at all. A WER near 19% therefore describes new recordings from airports the model knows; for a new airport, 26% is the better estimate, and a new region can be far worse. The Sion model also trained on the fewest clips (553), because Sion supplies the most.

Table 4. Leave one airport out: word error rate on each airport's clips for a model that never heard that airport, for the cross-validation model that heard other recordings from it, and for the unmodified model (beam search with repetition guards).

Airport left out	Clips	Held-out airport	Seen airport	Not fine-tuned
Prague (LKPR)	104	30.24%	24.85%	60.70%
Brno (LKTB)	32	22.57%	23.67%	57.74%
Sion (LSGS)	258	31.81%	20.00%	62.08%
Bern (LSZB)	173	17.69%	16.29%	44.03%
Zurich (LSZH)	126	20.74%	16.54%	48.67%
Bratislava (LZIB)	79	23.85%	19.08%	46.53%
Sydney (YSSY)	102	41.44%	18.80%	56.13%
All 874 clips	874	26.48%	19.16%	53.44%

5.3 Noise robustness

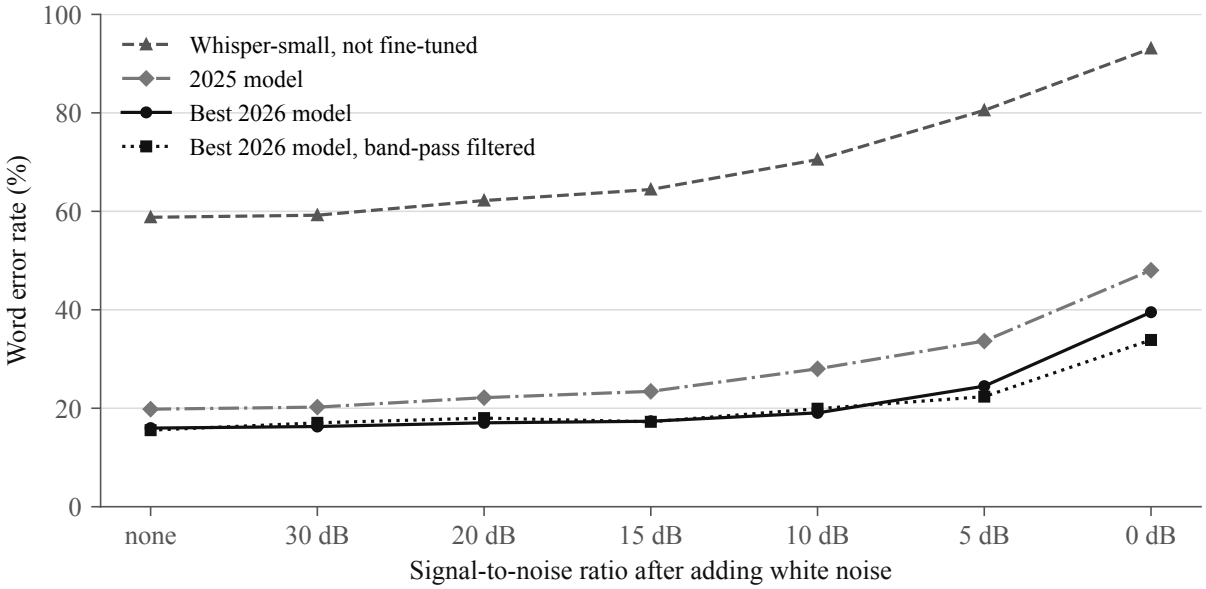


Figure 4. Word error rate on the 74 test clips as white noise is added. Greedy decoding with repetition guards. The filtered clips receive about 2 dB less noise at each nominal ratio, because the filter lowers the clip power the noise is set from.

Table 5. Word error rate as white noise is added.

Added noise	Best model, 2026	2025 model	Not fine-tuned
none	15.97%	19.81%	58.79%
20 dB	17.04%	22.15%	62.19%
10 dB	19.06%	28.01%	70.50%
0 dB	39.51%	48.03%	93.08%

The fine-tuned models are better than the unmodified one at every noise level, and the selected Whisper-medium soup loses fewer points than the 2025 model: 3.09 at 10 dB where the 2025 model loses 8.20, and 23.54 at 0 dB against 28.22. As a multiple of its starting error, however, it does not degrade more slowly at 0 dB (2.47 times against 2.42). The Whisper-small soup, trained on the same additional speech, follows the 2025 curve closely (45.79% at 0 dB, against 42.81% for the single Whisper-medium model with the same extra speech), which suggests that the larger model, rather than the extra data, accounts for most of the added robustness. Without the repetition guards the curves were not monotonic, because greedy decoding looped on a few noisy clips.

Applying a 300 to 3,400 Hz filter before adding noise barely changes the recordings as they are (15.55% instead of 15.97% for the selected model), because they have already passed through a radio channel. Under heavy noise the filtered versions score better (33.87% against 39.51% at 0 dB), but this is an artefact of the definition. The noise level is set from the clip's total power, and the filter removes a median 2.05 dB of each test clip's power (mean 2.28 dB), so a filtered clip receives about 2 dB less noise at the same nominal SNR while its speech in the band is almost unchanged. The noise itself is unfiltered in both cases, so the same share of it falls in the voice band either way. Near 0 dB the selected model's error rises by roughly 3.0 points per dB, so 2 dB less noise would account for about 6 points, as much as the whole 5.64-point gap: there is no sign that the filter itself helps.

5.4 From words to instructions

Table 6. Reference-transcript extraction agreement: clips in which a field the tagger extracts from the model's transcript matches the one it extracts from the correct transcript. A clip counts for a field only if the reference has that field.

Field	Not fine-tuned	2025 model	Best model, 2026
Callsign (59 clips)	4	35	38
Command (56 clips)	19	37	44
Numbers (58 clips)	12	35	46
Callsign, command and numbers all match (74 clips)	3	29	38
Every field, including waypoints (74 clips)	0	20	25

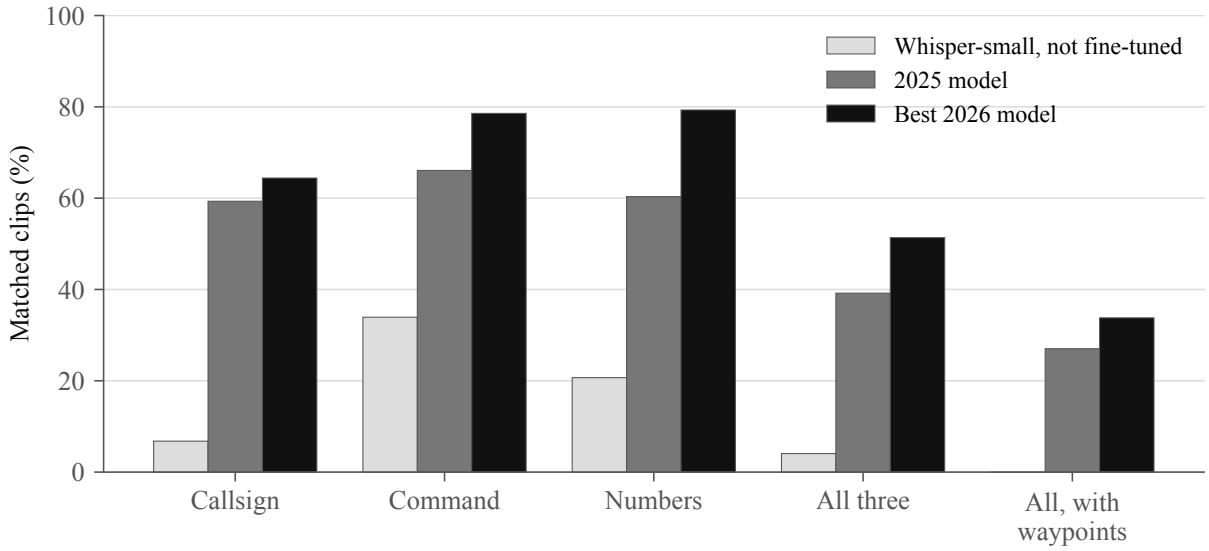


Figure 5. Share of test clips in which each extracted field matches the one extracted from the correct transcript.

A WER near 14% sounds like six words in seven being right, but a single wrong word can change who an instruction is for or what number it carries. Even the best model extracts the same callsign, commands and numbers as the correct transcript in only 38 of the 74 clips (Table 6). Waypoints are left out of that count because their labels come from a rule known to be too loose; with them, only 25 clips match in every field. These counts compare the fields as unordered collections; requiring the same order as in the correct transcript, which is closer to the same instruction for the same aircraft, gives the same counts for every model. Because the tagger is compared with itself, this is agreement, not accuracy: it measures how much transcription error reaches the extracted fields, not whether a person would read the instruction the same way. Spelling variants of the same word (alfa and alpha, oskar and oscar) are treated as the same, as in Section 5.5. The best model gets the most, but only just: the single Whisper-medium model and the Whisper-small soup both get 37, and differences of a few clips out of 74 are within chance. Among callsign, command and numbers, callsigns are the weakest field. The best model misses 24 of the 71 callsigns in the 59 clips that contain one, in 21 clips. None begins with a word absent from the training transcripts. Sorted by cause, by aligning each missed callsign with the stretch of the model's transcript that matches it best: 14 are phonetic-alphabet letters or number words heard wrong, missed or added, such as "hotel delta lima" heard as "hotel golf lima"; 7 were transcribed word for word but labelled differently by the tagger; 1 is a misheard airline name; and 2 are mostly lost. The 2025 model's 25 misses split 15, 6, 4 and 0. Letters and digits are the confusions the phonetic alphabet exists to prevent and that a noisy channel most easily causes.

5.5 Callsign letters and digits

Callsigns and registrations are spelled with the phonetic alphabet and digit words, so those words were scored on their own. Each reference letter or digit word is aligned with the model's transcript and counts as right if it is transcribed exactly; spelling variants of the same letter or digit (alfa and alpha, oskar and oscar, niner and nine) count as the same word, because the ATCO2 transcripts use both and UWB-ATCC always writes "nine".

Table 7. Spelled letters and digit words transcribed exactly (479 on validation, 429 on test).

Model	Validation	Test
Whisper-small, not fine-tuned	60.96%	66.20%
2025 model	91.23%	93.24%
Best model, 2026	96.24%	94.64%

Individual letters and digits are therefore mostly right; the remaining errors are scattered confusions such as delta heard as golf or papa as bravo. A callsign fails as a whole when any one of its words is wrong, which is why whole callsigns match far less often than their letters. Two attempts to improve letters were made on the best model at the time, the 6-epoch soup (96.03% of letters and digits on validation). Giving Whisper the phonetic alphabet as a prompt, under its best decoding on validation (plain beam), made no measurable difference to WER (171 word errors against 170) and got slightly fewer letters right (94.27% against 95.54%), so it was not adopted; under guarded beam it did much worse (16.96%), possibly because the repetition guards also count the prompt's words. A second attempt trained that model for two more epochs with fresh adapters and a loss that counts letter and digit words three times as heavily as other words. The rule was fixed before the run: an epoch would be kept only if it beat that model on validation for both letters and digits and overall WER. After one epoch it got 96.87% of letters and digits right against 96.03%, only four more of 479 words, while its WER rose from 14.86% to 15.38%; after two epochs the letters gain shrank (96.45%) and WER rose to 16.00%. Neither was adopted or scored on test.

5.6 The word tagger

Against its rule-generated references, the application's DistilBERT tagger [9] agrees with the rules on 871 of 1,001 words, a weighted F1 of 0.8657. These are the same sentences its training used to pick its best epoch, so the figure is somewhat optimistic even as a measure of agreement. A separate BERT tagger [10] scored 0.9544, but against a version of the rules that disagrees on 25 of the words, so the two cannot be compared. The largest error group is 66 of the 140 words the rules call ordinary being tagged as waypoints (Figure 6). Because the references come from rules, these scores measure agreement with the rules, not accuracy. The rules also call "and" a waypoint, and so does the tagger (Section 6).

The tagger has not yet been scored against labels written by a person. As a stand-in, the 1,001 words were labelled by an AI model following a written guide fixed before any labelling, reading only the sentences; this is not a human evaluation, and the guide's conventions (for example, that "runway" belongs to the number after it) differ in places from the rules'. Leaving out the 19 words it marked unsure, the tagger agrees with those labels on 57.33% of words (weighted F1 0.6233), and the rules themselves on 65.07% (0.7011). Most of the disagreement is the known waypoint problem: of the 247 words labelled ordinary, the tagger calls 173 waypoints. Another 71 callsign words, mostly flight-number digits, are tagged as values. The application's actual output, after it drops waypoint labels on function words, agrees on 60.18% (weighted F1 0.6532). Whole entities match exactly far less often: for callsigns, 51.43% of the application's callsign spans are exactly right and it finds 45.57% of the labelled ones. The 0.8657 therefore overstates the tagger's accuracy; a person's labels are still needed to measure it, and because these 89 sentences also shaped the tagger's development, a fixed or retrained tagger would need new sentences for its final evaluation.

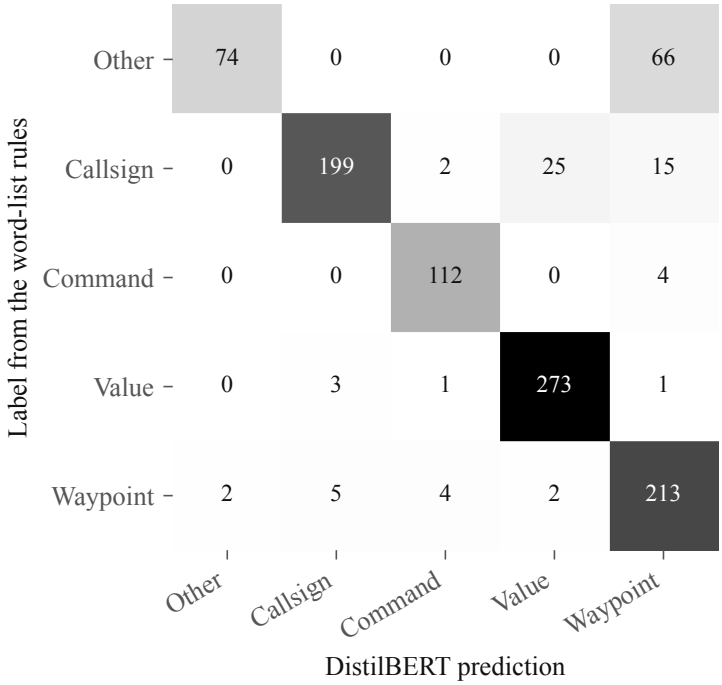


Figure 6. DistilBERT confusion matrix on 1,001 words, against labels from the word-list rules.

6. Discussion

For context, van Doorn et al. [12] fine-tuned Whisper on several ATC corpora and report 13.5% WER on ATCO2. Their data and split differ from those here, so the two numbers cannot be compared directly. Their ATCOSIM result, 1.17% with a random split against 3.88% when speakers are held out, shows the same effect that the audit in Section 3 found: letting recordings or speakers appear on both sides of a split flatters the score.

The main result is that, on recordings the model has never heard, larger models with low-rank adapters reduce WER from 18.64% to 14.27% when two of them are averaged, that this gain survives a paired test on 74 clips, and that the ablation attributes most of it to the larger model rather than the extra speech. Most of the other interventions did not produce gains that a test of this size can detect, and one of them, synthetic radio speech, pointed the wrong way in these experiments, which does not show that synthetic ATC speech cannot help. The synthetic clips were designed with care, but text-to-speech voices are calm and clear where controllers are fast and clipped; a simulated channel changes the sound quality but not the way people speak. Real speech from another corpus, by contrast, made the model much less prone to looping (UWB-ATCC is from Czech airspace, which the ATCO2 subset also covers, through Prague and Brno), while the larger model accounted for most of the gain in accuracy and in robustness to noise.

The measured improvement is modest next to the effect of fine-tuning itself, and the instruction-level results show why it matters anyway: at this level of WER most instructions still lose at least one field. For uses such as filling in radar labels, the relevant measure is whether each command, value and callsign is right [8], and on that measure callsigns, and within them spelled-out letters, are where further work should go.

The application reflects these results. It now uses the 2026 Whisper-medium soup when it is installed, runs it on an Apple or NVIDIA GPU when one is available, and decodes with guarded beam search. On the example recording used throughout the project, which says “push and start approved”, the 2025 model produced “portion startup approved”; the 2026 model produces “provision start approved”, still wrong, though it gets the callsign and squawk code in the same call right. The tagger then labels each “and” in the transcript as a waypoint, a habit it learned from the word-list rules, which do the same; the application now removes waypoint labels from English function words, but the underlying fix is better rules and human-checked labels.

Limitations. The test set has 74 clips from 66 recordings; it can confirm large effects but not separate models a point or two apart. Cross-validation over all 874 clips narrows the estimate for the Whisper-small

recipe, but was not affordable for every configuration. Each configuration was trained once, so the effect of random initialisation and data order is not measured. The 2025 model's checkpoint was chosen with data that included the test clips. The ablation separates model size from the extra speech only roughly, because it also trained on less audio. The validation clips come from training recordings, so choosing on them can reward memorising those recordings: run E is worse than run A on validation but better on test. Validation ties between decodings went to guarded beam, the first tried. Run E and the two 6-epoch Whisper-medium runs were still improving when their epoch budgets ran out; the 12-epoch repeat peaked at epoch 9. The best model differs from the 2025 model in size, training data and recipe at once, so its gain is not the effect of the fine-tuning technique alone, and the cross-validation and airport results describe the Whisper-small recipe, not the best model. Some ideas were developed after earlier test results were known (Section 4.4). "Unseen" means a different recording, not a different airport or controller; Table 4 shows what a new airport costs. The tagger has never been evaluated against human labels. The unmodified model is penalised slightly by formatting differences that spelling out digits does not remove. No test involved live traffic, and nothing here is fit for operational use.

7. Conclusion

With one hour of real audio, fine-tuning cut Whisper-small's word error rate on unseen ATC recordings by two thirds, from 56.87% to 18.64%. Adding 10.5 hours of a second corpus and moving to Whisper-medium brought it to 14.27%. After an audit that separated recordings, moved every decision onto a validation set and attached intervals to every comparison, the best model, an average of two Whisper-medium models trained with LoRA adapters, reached 14.27%, a gain of 4.37 points that the test can confirm, together with better robustness to noise; most of the gain comes from the larger model. The test clips are disjoint from training but not untouched, since some later ideas followed earlier test results. Cross-validation over every recording confirms the Whisper-small recipe at 19.16%. Synthetic speech did not help in these experiments. The next steps are human labels for the tagger, more unseen test recordings, and direct work on the recognition of callsign letters and digits. A full log of every experiment, including those that failed, is in `docs/experiment_log.md`.

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Data and code availability

Code, result summaries and the scripts that produce every table and figure are at https://github.com/Sinecep tor/ATCO. Results are in `results/`, and each number in this paper can be recomputed from them. Model weights and audio are not redistributed: the ATCO2 subset is available from https://www.atco2.org/data and UWB-ATCC from https://huggingface.co/datasets/Jzuluaga/uwb_atcc.

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